

KALYAN KUMAR REDDY BONTHA

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EDUCATION

Master of Science in Computer Science

Aug 2023 – May 2025

Texas A&M University, Corpus Christi, Texas

Relevant Coursework: AI, Deep Learning, Network Security, HCI.

Bachelor of Computer Science and Engineering

Aug 2018 – Jul 2022

CMR University, Bangalore, India

Relevant Coursework: Cloud Computing, DBMS, Mobile Application Development, AI, Deep Learning, Python, Web Design

SKILLS

- **Programming:** Python, C, HTML, CSS, Boot Strap, PHP, ML.
- **AI/ML Frameworks:** NumPy, Pandas, TensorFlow
- **Developer Tools:** Visual Studio, Python Idle, Jupyter Notebook, Putty, Wamp Server.
- **Database/ Services:** MySQL, SQL lite, Amazon Web Services.

PROJECTS

UAV Audio Detection

May 2025

- Set new benchmarks in drone acoustic recognition, advancing the field with multi-class classification and novel pitch-shifted sample techniques.
- Developed a high-accuracy drone classification system (97.23% F1-score) using CNN, RNN, CRNN on Mel & STFT spectrograms, identifying Bebop, Mambo, and Phantom drones.
- Proven Mel spectrogram efficiency, outperforming STFT in drone classification by enhancing perceptual features and computational efficiency.
- Augmented dataset with 3000+ samples, integrating pitch shifting, YouTube recordings, and online databases for robust model generalization.
- Implemented entropy thresholding for unknown drone detection, achieving 80% accuracy in identifying unauthorized UAVs.
- Optimized deep learning architectures, reducing parameters for real-time detection, and achieving 96.74% accuracy with CNN-Mel models.

Health Care Chatbot-AI - Capstone Project | PHP, MySQL, WAMP, HTML/CSS.

March 2022

- Designed and developed an AI-powered chatbot system to assist users with healthcare queries and 24/7 medical support.
- Implemented user and admin modules with features like appointment assistance, doctor lookup, and automated Q&A using natural language processing techniques.
- Created structured databases and dynamic UI using PHP and MySQL, deployed locally via WAMP Server.
- Ensured system usability through ER diagrams, UML models (Use Case, Activity, Sequence), and tested for real-time user interactions.
- Emphasized secure data handling, scalability, and efficient response generation for various patient inputs.

ACHIEVEMENTS

- Awarded Merit Scholarship at Texas A&M University-Corpus Christi for outstanding academic and extracurricular activities on campus.
- Received offer from Wipro ltd. As project engineer during final year of Under Graduation.
- Certificate of Achievement for the Distinguished Capstone project "UAV Audio Detection".

CERTIFICATIONS

- **AWS Certified Cloud Practitioner**– Proven knowledge of AWS cloud services and architecture